Jackie Baek

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EMPLOYMENT

New York University, Stern School of Business 2023 -
Assistant Professor, Technology, Operations & Statistics

Simons Institute for the Theory of Computing, UC Berkeley Fall 2022
Research Fellow in the program *Data-Driven Decision Processes*

EDUCATION

Massachusetts Institute of Technology 2016 - 2022
Ph.D. in Operations Research
Thesis: Decision-Making Under Uncertainty: From Theory to Practice
Advisor: Vivek F. Farias

University of Waterloo 2011 - 2016
Bachelor of Mathematics
Joint Honours Computer Science & Combinatorics and Optimization

JOURNAL PUBLICATIONS

The Feedback Loop of Statistical Discrimination
Jackie Baek, Ali Makhdoumi
Management Science (forthcoming)
– Appeared in the *ACM Conference on Equity and Access in Algorithms, Mechanisms, and Optimization (EAAMO), 2025*

Leveraging Reusability: Improved Competitive Ratio of Greedy for Reusable Resources
Jackie Baek, Shixin Wang
Operations Research Letters, 2026

Policy Optimization for Personalized Interventions in Behavioral Health
Jackie Baek, Justin J. Boutilier, Vivek F. Farias, Jónas Oddur Jónasson, Erez Yoeli
Manufacturing and Service Operations Management, 2025
– Winner, *Pierskalla Best Paper Award 2024*

The Limits to Learning a Diffusion Model
Jackie Baek, Vivek F. Farias, Andreea Georgescu, Retsef Levi, Tianyi Peng, Deeksha Sinha, Joshua Wilde, Andrew Zheng
Management Science, 2025
– Appeared in the *22nd ACM conference on Economics and Computation, 2021*

Fair Exploration via Axiomatic Bargaining
Jackie Baek, Vivek F. Farias
Management Science, 2024
– Appeared in *NeurIPS 2021 (Spotlight, top 3% of submissions)*
– Second Place, *MSOM Student Paper Competition 2022*
– Finalist, *George Nicholson Student Paper Competition 2021*
– Finalist, *RMP Jeff McGill Student Paper Award 2021*
– Honorable Mention, *MIT ORC Best Student Paper Competition 2021*
– Oral presentation, *ACM Conference on Equity & Access in Algorithms, Mechanisms, & Optimization (EAAMO), 2021*

Technical Note–Bifurcating Constraints to Improve Approximation Ratios

for Network Revenue Management with Reusable Resources
Jackie Baek, Will Ma
Operations Research, 2022
Appeared in the *12th International Symposium on Algorithmic Game Theory, 2019*

Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the United States
COVID-19 Forecast Hub (298 authors)
Proceedings of the National Academy of Sciences, 2022
– This paper resulted from contributing COVID-19 forecasts (from the paper "The Limits to Learning a Diffusion Model") to the COVID-19 Forecast Hub.

A Game-Theoretic Analysis of Reallocation Mechanisms for Airport Landing Slots
Jackie Baek, Hamsa Balakrishnan
IEEE Transactions on Intelligent Transportation Systems, 2020

Conference Publications (no journal version)

When Collaborative Filtering is not Collaborative: Unfairness of PCA for Recommendations
David Liu, Jackie Baek, Tina Eliassi-Rad
ACM Conference on Fairness, Accountability, and Transparency (FAccT) 2025

TS-UCB: Improving on Thompson Sampling With Little to No Additional Computation
Jackie Baek, Vivek F. Farias
International Conference on Artificial Intelligence and Statistics (AISTATS) 2023

Working Papers

Social Learning with Limited Attention: Negative Reviews Persist under Newest First
Jackie Baek, Atanas Dinev, Thodoris Lykouris
Major revision, *Operations Research*
– Appeared in the *25th ACM conference on Economics and Computation, 2024*
– Second Place, *Junior Faculty Interest Group (JFIG) Paper Competition, 2025*

Strategic Hiring under Algorithmic Monoculture
Jackie Baek, Hamsa Bastani, Shihan Chen

Personalized Promotions in Practice: Dynamic Allocation and Reference Effects
Jackie Baek, Will Ma, Dmitry Mitrofanov

Balancing Customer Engagement and Annoyance in Online Retail: Insights from a Field Experiment
Jackie Baek, Daniel Chen, Will Ma, Dmitry Mitrofanov
– Appeared in the *27th ACM Conference on Economics and Computation, 2026*

AI Agents for Inventory Control: Human-LLM-OR Complementarity
Jackie Baek, Yaopeng Fu, Will Ma, Tianyi Peng
Major revision, *Management Science*
– Appeared in the *27th ACM Conference on Economics and Computation, 2026*

LLM-SAA: LLM-persona Generated Distributions for Decision-making
Jackie Baek, Yunhan Chen, Ziyu Chi, Will Ma

Optimal Exploration of New Products under Assortment Decisions
Jackie Baek, Atanas Dinev, Thodoris Lykouris
Major revision, *Management Science*

Misspecified Estimate-then-Optimize Leads to Supra-Competitive Prices
Jackie Baek, Vivek F. Farias, Farrell Wu
Major revision, *Management Science*

LLMs Can Design Near-Optimal OR Algorithms
Jackie Baek

Teaching

NYU Stern

Instructor, Operations Management (Undergraduate Core)	2024, 2025
Instructor, Advanced Topics in Operations for Societal Impact (PhD)	2025
Guest Lecture for PhD class	2022, 2024

Cornell Tech

Guest Lecture for PhD class (INFO 6455)	2025

Columbia University

Guest Lecture for PhD class (IEOR 8100)	2024

Wharton School, University of Pennsylvania

Guest Lecture for PhD class	2022

MIT Sloan

Teaching Assistant, Operations Management (MBA Sloan Fellows)	2020
Teaching Assistant, The Analytics Edge (MBA)	2018
Instructor (one lecture), Computing in Optimization and Statistics	2017, 2018

Invited Talks

2026: Simons Workshop on Bridging Prediction and Intervention Problems in Social Systems, Duke Fuqua OM, Berkeley Haas

2025: BIRS Workshop on Dynamic Allocation and Matching, OM in the Wild Workshop, CMU Tepper

2024: BIRS Workshop on Combinatorial Optimization for Online Platforms, NYU Langone HiBRID Cafe, TTIC Workshop on Data-Driven Decision Processes, NYU Theory CS Seminar

2023: CUHK Business School, IMSI Workshop on Analytics for Improved Healthcare, NYU Digital Health Research Workgroup, UMass Amherst CS Theory

2022: Northwestern Kellogg, Columbia IEOR, USC Marshall, Johns Hopkins Carey, NYU Stern, Stanford GSB, Duke Fuqua DS, Yale SOM, Michigan Ross, Chicago Booth, UToronto Rotman Young Scholar Seminar, Caltech RSRG, LinkedIn Responsible AI, Workshop on Information and Learning

2021: UPenn Wharton, UBC Sauder, UNC Kenan-Flagler, Cornell ORIE, MIT OM Seminar, MIT Data Science Lab Seminar, Cornell Young Researchers Workshop

Service

Reviewer for Journals: *Management Science, Operations Research, Manufacturing & Service Operations Management, Operations Research Letters, Mathematics of Operations Research, European Journal of Operational Research, Journal of Machine Learning Research, IEEE Control Systems Letters, IEEE Transactions on Intelligent Transportation Systems, INFORMS Journal on Computing, Health Care Management Science, MIS Quarterly*

Program Committee/Reviewer for Conferences: *EC 2023-2025, FAccT 2022-2025, EAAMO 2023-2025, WINE 2022, ALT 2023, The Web Conference 2023, AISTATS 2023, NeurIPS 2023, ICLR 2024, COLT 2024, MSOM SIG 2024-2025, RMP Student Paper Competition 2024, MSOM Student Paper Competition 2025*

Co-chair, Pierskalla Best Paper Award 2025

Co-organizer, TTIC Summer Workshop on *Data-Driven Decision Processes: From Theory to Practice* 2024

Co-organizer, NYC Operations Day 2024-2026

Organizer, Stern OM Seminar Series 2023-2025

Member, Stern OM Faculty Recruiting Committee 2023, 2024

Member, Stern OM PhD Admissions Committee 2024, 2025

Honors and Awards

Second Place, Junior Faculty Interest Group (JFIG) Paper Competition 2025
Winner, Pierskalla Best Paper Award 2024
Operations Research Meritorious Service Award 2025
Management Science Distinguished Service Award 2023, 2025
Management Science Meritorious Service Award 2024
Second Place, MSOM Student Paper Competition 2022
Finalist, George Nicholson Student Paper Competition 2021
Finalist, RMP Jeff McGill Student Paper Award 2021
Honorable Mention, MIT ORC Best Student Paper Competition 2021
Finalist, Post-Pandemic Supply Chain and Healthcare Management Best Paper Competition 2021
Runner-up, MIT LIDS Student Conference Best Presentation 2018
NSERC Undergraduate Student Research Award 2015
Mathematics National Scholarship, University of Waterloo 2011 - 2016

Work Experience

GRAIL Palo Alto, CA
Machine Learning Engineer Intern Summer 2018
Investigated genomic features on its ability to improve detecting early-stage cancer

Snap Venice, CA
Software Engineer Intern Fall 2014, Summer 2016
Improved app startup performance by implementing incremental updates

Bloomberg London, UK
Software Engineer Intern Fall 2015
Optimized a financial dashboard to minimize redundant function calls

Dropbox San Francisco, CA

Software Engineer Intern Fall 2013, Spring 2014
Optimized sync by implementing delta compression using finite-state machines

LogicBlox Atlanta, GA
Software Engineer Intern Spring 2013